

Concentration Units

Answer key with worked solutions

A. Ideas

1. Molarity and molality.

(a) **Molarity** is moles of solute per **litre of solution**; **molality** is moles of solute per **kilogram of solvent**. (b) **Molarity** changes with temperature, because the volume of a solution expands or contracts. Molality, mass percent, ppm and mole fraction use **masses** and amounts, which do not change.

B. Mass-based units

2. Sugar.

$$\text{Mass of solution} = 12.0 \text{ g} + 188 \text{ g} = 200. \text{ g}; \frac{12.0 \text{ g}}{200. \text{ g}} \times 100\% = \mathbf{6.00\%}$$

3. Fluoride.

$$0.45 \text{ mg} = 0.00045 \text{ g}; \frac{0.00045 \text{ g}}{500. \text{ g}} \times 10^6 = \mathbf{0.90 \text{ ppm}}$$

4. Saline.

$$0.0090 \times 500. \text{ g} = \mathbf{4.5 \text{ g NaCl}}$$

C. Moles-based units

5. Glucose.

$$\begin{aligned} \text{Moles: glucose } & \frac{18.0 \text{ g}}{180.16 \text{ g/mol}} = 0.09991 \text{ mol}; \text{ water } \frac{90.0 \text{ g}}{18.02 \text{ g/mol}} = 4.994 \text{ mol} \\ \text{(a) } x &= \frac{0.09991 \text{ mol}}{0.09991 \text{ mol} + 4.994 \text{ mol}} = \mathbf{0.0196} \\ \text{(b) } m &= \frac{0.09991 \text{ mol}}{0.0900 \text{ kg}} = \mathbf{1.11 \text{ mol/kg}} \end{aligned}$$

D. Converting between units

6. Sulfuric acid.

Take 1.000 L: mass = 1000 mL $\times$ 1.84 g/mL = 1840 g

Mass of H₂SO₄ = 0.980 $\times$ 1840 g = 1803 g; moles = $\frac{1803 \text{ g}}{98.08 \text{ g/mol}} = 18.38 \text{ mol}$

$$c = \frac{18.38 \text{ mol}}{1.000 \text{ L}} = \mathbf{18.4 \text{ mol/L}}$$

7. From molality to mass percent.

Take 1.000 kg of water: it contains 1.50 mol NaCl = 1.50 mol $\times$ 58.44 g/mol = 87.66 g.

$$\begin{aligned}\% \text{ NaCl} &= \frac{87.66 \text{ g}}{1000 \text{ g} + 87.66 \text{ g}} \times 100\% \\ &= \mathbf{8.06\%}\end{aligned}$$

E. Challenge

8. From molality to molarity.

Take 1.000 kg of water: 3.00 mol NaCl = 3.00 mol $\times$ 58.44 g/mol = 175.3 g.

Mass of solution = 1000 g + 175.3 g = 1175.3 g

Volume of solution = $\frac{1175.3 \text{ g}}{1.12 \text{ g/mL}} = 1049 \text{ mL} = 1.049 \text{ L}$

$$c = \frac{3.00 \text{ mol}}{1.049 \text{ L}} = \mathbf{2.86 \text{ mol/L}}$$